Reference

Reading Hangul Block by Block: A Quick Reference

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Abstract—A quick reference for reading Hangul, the Korean writing system, covering the structure of syllable blocks and the pronunciation of all consonants and vowels.

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1. INTRODUCTION

Hangul is not a logographic script like Chinese characters, but an alphabet of 14 basic consonants and 10 basic vowels, created in 1443 by King Sejong the Great to make literacy accessible to everyone [1]. The consonants mimic the shape of the speech organs that produce them, and the vowels are built from three symbols representing heaven(·), earth(ㅡ), and human(ㅣ) [1]. Letters are not written in a line, but grouped into blocks, one per syllable. Learning to read these blocks is the key to reading Hangul.

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2. STRUCTURE

Every syllable is written as a single block built from up to three parts: an initial consonant c_i, a total vowel v_t, and a final optional consonant c_o.

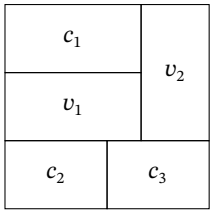


Figure 1. Basic structure of a Hangul syllable block

fig. 1 shows the block structure. Letters are read in this order:

$$\text{Syllable} = \underbrace{c_1}_{\text{initial consonant } c_i} + \underbrace{(v_1 + v_2)}_{\text{total vowel } v_t} + \underbrace{(c_2 + c_3)}_{\text{optional consonant } c_o} \tag{1}$$

c_o eq. (1) If the optional consonant c_o uses both c_2 and c_3, then they are spoken together as a single consonant sound by compressing the sound of the individual consonants.

3. CONSONANTS

The consonants are grouped by the speech organ that produces them. Each consonant is shown in this notation:

$$\begin{pmatrix} \textbf{hangul letter} & & \\ \text{sound}^{\text{default}}_{\text{reference word}} & \text{sound}^{\text{special case Nr.}}_{\text{reference word}}(\text{trigger}) & \text{as } \mathbf{c_1} \\ \text{sound}^{\text{default}}_{\text{reference word}} & \text{sound}^{\text{special case Nr.}}_{\text{reference word}}(\text{trigger}) & \text{as } \mathbf{c_2} \\ \text{sound}^{\text{default}}_{\text{reference word}} & \text{sound}^{\text{special case Nr.}}_{\text{reference word}}(\text{trigger}) & \text{as } \mathbf{c_3} \end{pmatrix} \tag{2}$$

Pronunciations are simplified for beginners, based on the official rules [2], and written as an English speaker would read them rather than in the official romanization (e.g. ㅡ as “eww” instead of “eu”). Onomatopoeia and interjections are marked with quotes, ∅ means silent, and numbered special cases are listed below each bracket.

3.1. Tongue, velar

Back of tongue against the soft palate; ㄱ shows the tongue root.

$$\begin{pmatrix} \text{ㄱ} \\ \text{g}_{\mathbf{go}} \\ \text{g}_{\mathbf{go}} \\ \text{g}_{\mathbf{go}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅋ} \\ \text{k}_{\mathbf{k}\text{ite}} \\ \text{g}_{\mathbf{go}} \\ - \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㄲ} \\ \text{gg}_{\text{“}\mathbf{G}\text{ulp!”}} \\ - \\ - \end{pmatrix} \tag{3}$$

3.2. Tongue, alveolar

Tongue tip touches the ridge behind the upper teeth; ㄴ shows the tongue.

$$\begin{pmatrix} \text{ㄴ} \\ \text{n}_{\mathbf{no}} \\ \text{n}_{\mathbf{no}} \\ - \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㄷ} \\ \text{d}_{\mathbf{do}} \\ \text{d}_{\mathbf{do}} \\ - \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅌ} \\ \text{t}_{\text{top}} \\ \text{d}_{\mathbf{do}} \\ \text{t}_{\text{top}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㄸ} \\ \text{dd}_{\text{“}\mathbf{D}\text{ing!”}} \\ - \\ - \end{pmatrix} \tag{4}$$

$$\begin{pmatrix} \text{ㄹ} \\ \text{r}_{\text{Maria}} \quad \text{l}^{1}_{\textbf{ball}}(c^{\text{prev}}_{end} = \text{ㄹ}) \\ \text{l}_{\text{ball}} \\ - \end{pmatrix} \tag{5}$$

ㄹ eq. (5) special cases:

1. **If previous syllable ends in ㄹ**, then ㄹ in $\mathbf{c}_1$ is read as **l not r**. (빨리 → bbal-**r**i ✗, bbal-**l**i ✓).

3.3. Lips, bilabial

Lips pressed together; ㅁ shows the mouth.

$$\begin{pmatrix} \text{ㅁ} \\ \text{m}_{\textbf{m}\text{oon}} \\ \text{m}_{\textbf{m}\text{oon}} \\ \text{m}_{\textbf{m}\text{oon}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅂ} \\ \text{b}_{\textbf{b}\text{oy}} \\ \text{b}_{\textbf{b}\text{oy}} \\ \text{b}_{\textbf{b}\text{oy}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅍ} \\ \text{p}_{\textbf{p}\text{ie}} \\ \text{b}_{\textbf{b}\text{oy}} \\ \text{b}_{\textbf{b}\text{oy}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅃ} \\ \text{bb}_{\text{“}\textbf{B}\text{ang!”}} \\ - \\ - \end{pmatrix} \tag{6}$$

3.4. Teeth, sibilant

Air hisses past the teeth; ㅅ shows the gap between teeth.

$$\begin{pmatrix} \text{ㅅ} \\ \text{s}_{\textbf{s}\text{un}} \quad \text{sh}^{1}_{\textbf{sh}\text{e}}(\text{v}_t = \text{ㅣ}) \\ \text{d}_{\textbf{d}\text{o}} \quad \text{s}^{2}_{\textbf{s}\text{un}}(c^{\text{next}}_{1} = \text{ㅇ}) \\ \text{d}_{\textbf{d}\text{o}} \quad \text{s}^{2}_{\textbf{s}\text{un}}(c^{\text{next}}_{1} = \text{ㅇ}) \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅆ} \\ \text{ss}_{\textbf{s}\text{ea}} \\ - \\ - \end{pmatrix} \tag{7}$$

ㅅ eq. (7) special cases:

1. **If v_1 is empty and v_2 is ㅣ**, then ㅅ in $\mathbf{c}_1$is read as **sh not s** (시 → **s**i ✗, **sh**i ✓).
2. **If the next syllable starts with ㅇ**, then any ㅅ in $\mathbf{c}_2$ and $\mathbf{c}_3$ is read as **s not d**.
   (옷이 → o**d**-i ✗, o**s**-i ✓)
   (있어 → id**d**-uh ✗, is**s**-uh ✓).

$$\begin{pmatrix} \text{ㅈ} \\ \text{j}_{\textbf{j}\text{am}} \\ \text{d}_{\textbf{d}\text{o}} \quad \text{j}^{1}_{\textbf{j}\text{am}}(c^{\text{next}}_{1} = \text{ㅇ}) \\ \text{d}_{\textbf{d}\text{o}} \quad \text{j}^{1}_{\textbf{j}\text{am}}(c^{\text{next}}_{1} = \text{ㅇ}) \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅊ} \\ \text{ch}_{\textbf{ch}\text{ild}} \\ \text{d}_{\textbf{d}\text{o}} \\ - \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅉ} \\ \text{jj}_{\text{Gu}\textbf{cc}\text{i}} \\ - \\ - \end{pmatrix} \tag{8}$$

ㅈ eq. (8) special cases:

1. **If the next syllable starts with ㅇ**, then any ㅈ in $\mathbf{c}_2$ and $\mathbf{c}_3$ is read as **j not d**.
   (맞아 → ma**d**-a ✗, ma**j**-a ✓)
   (앉아 → an**d**-a ✗, an**j**-a ✓)

3.5. Throat, glottal

Sound from the throat; ㅇ shows the open throat

$$\begin{pmatrix} \text{ㅇ} \\ \varnothing \\ \text{ng}_{\textbf{sing}} \\ - \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅎ} \\ \text{h}_{\textbf{h}\text{at}} \\ \varnothing \\ \varnothing \end{pmatrix} \tag{9}$$

4. VOWELS

The vowels are built from ·, ㅡ, and ㅣ (see the introduction). For reading, it is easiest to sort them by where they sit in the block: horizontal (v_1), vertical (v_2), and compound ($\text{v}_1 + \text{v}_2$). Each vowel is shown in this notation:

$$\begin{pmatrix} \text{hangul letter} \\ \text{vowel}_{\text{reference word}} \end{pmatrix} \tag{10}$$

4.1. Horizontal Vowels v_1

Based on ㅡ, written below c_1.

$$\begin{pmatrix} \text{ㅡ} \\ \text{eww}_{\text{“}\textbf{eww}\text{!”}} \end{pmatrix} \tag{11}$$

$$\begin{pmatrix} \text{ㅗ} \\ \text{o}_{\textbf{go}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅛ} \\ \text{yo}_{\textbf{yo}\text{ga}} \end{pmatrix} \tag{12}$$

$$\begin{pmatrix} \text{ㅜ} \\ \text{oo}_{\textbf{moon}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅠ} \\ \text{you}_{\textbf{you}} \end{pmatrix} \tag{13}$$

4.2. Vertical Vowels v_2

Based on ㅣ, written to the right of c_1.

$$\begin{pmatrix} \text{ㅣ} \\ \text{i}_{\textbf{see}} \end{pmatrix} \tag{14}$$

$$\begin{pmatrix} \text{ㅏ} \\ \text{a}_{\text{spa}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅑ} \\ \text{ya}_{\textbf{yard}} \end{pmatrix} \tag{15}$$

$$\begin{pmatrix} \text{ㅓ} \\ \text{uh}_{\textbf{sun}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅕ} \\ \text{yuh}_{\textbf{you}\text{ng}} \end{pmatrix} \tag{16}$$

$$\begin{pmatrix} \text{ㅐ} \\ \text{ae}_{\textbf{ca}\text{t}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅒ} \\ \text{yae}_{\textbf{yea}\text{h}} \end{pmatrix} \tag{17}$$

$$\begin{pmatrix} \text{ㅔ} \\ \text{e}_{\textbf{he}\text{y}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅖ} \\ \text{ye}_{\textbf{ya}\text{y}} \end{pmatrix} \tag{18}$$

4.3. Compound vowels $\text{v}_1 + \text{v}_2$

A horizontal and a vertical vowel combined, filling both positions.

$$\begin{pmatrix} \text{ㅢ} \\ \text{ui}_{\text{“}\textbf{eww}\text{!”}+\textbf{ee}} \end{pmatrix} \tag{19}$$

$$\begin{pmatrix} \text{ㅚ} \\ \text{oe}_{\textbf{we}\textbf{t}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅘ} \\ \text{wa}_{\textbf{wa}\text{sabi}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅙ} \\ \text{wae}_{\textbf{wa}\text{gon}} \end{pmatrix} \tag{20}$$

$$\begin{pmatrix} \text{ㅟ} \\ \text{wi}_{\textbf{week}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅝ} \\ \text{wo}_{\textbf{wo}\text{nder}} \end{pmatrix} \bullet\!\!-\!\!\bullet \begin{pmatrix} \text{ㅞ} \\ \text{we}_{\textbf{we}\text{dding}} \end{pmatrix} \tag{21}$$

5. EXAMPLES

Each example breaks a syllable into its parts and then reads them in order, using this notation:

$$\begin{pmatrix} \textbf{hangul letter} \\ \boxed{\text{position in Block}} \\ \text{default sound} \\ \text{special case} \end{pmatrix} \tag{22}$$

5.1. Initial consonant + single vowel

$$\text{가} = \underbrace{\begin{pmatrix} \text{ㄱ} \\ \boxed{\text{c}_1} \\ \text{g} \end{pmatrix}}_{\text{c}_i} + \underbrace{\begin{pmatrix} \text{ㅏ} \\ \boxed{\text{v}_1} \quad \boxed{\text{v}_2} \\ \text{a} \end{pmatrix}}_{\text{v}_t} + \underbrace{\begin{pmatrix} \\ \boxed{\text{c}_2} \quad \boxed{\text{c}_3} \\ \end{pmatrix}}_{\text{c}_o} \tag{23}$$

$$= \begin{pmatrix} \text{ㄱ} \quad \text{ㅏ} \quad \\ \boxed{\text{c}_i} \quad \boxed{\text{v}_t} \quad \boxed{\text{c}_o} \\ \text{g} \quad \text{a} \end{pmatrix} = \begin{pmatrix} \text{가} \\ \boxed{\text{syllable}} \\ \text{ga} \end{pmatrix} \tag{24}$$

5.2. Initial consonant + compound vowel

$$\text{와} = \underbrace{\begin{pmatrix} \text{ㅇ} \\ \boxed{c_1} \\ \varnothing \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \text{ㅗ} & \text{ㅏ} \\ \boxed{v_1} & \boxed{v_2} \\ \text{o} & \text{a} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \boxed{c_2} & \boxed{c_3} \end{pmatrix}}_{c_o} \tag{27}$$

$$= \begin{pmatrix} \text{ㅇ} & \text{ㅘ} & \boxed{c_o} \\ \boxed{c_i} & \boxed{v_t} & \\ \varnothing & \text{wa} & \end{pmatrix} = \begin{pmatrix} \text{와} \\ \boxed{\text{syllable}} \\ \text{wa} \end{pmatrix} \tag{28}$$

5.3. Initial consonant + single vowel + optional consonant

$$\text{밥} = \underbrace{\begin{pmatrix} \text{ㅂ} \\ \boxed{c_1} \\ \text{b} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \text{ㅏ} \\ \boxed{v_1} & \boxed{v_2} \\ & \text{a} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \text{ㅂ} \\ \boxed{c_2} & \boxed{c_3} \\ \text{b} & \end{pmatrix}}_{c_o} \tag{29}$$

$$= \begin{pmatrix} \text{ㅂ} & \text{ㅏ} & \text{ㅂ} \\ \boxed{c_i} & \boxed{v_t} & \boxed{c_o} \\ \text{b} & \text{a} & \text{b} \end{pmatrix} = \begin{pmatrix} \text{밥} \\ \boxed{\text{syllable}} \\ \text{bab} \end{pmatrix} \tag{30}$$

$$\text{꽃} = \underbrace{\begin{pmatrix} \text{ㄲ} \\ \boxed{c_1} \\ \text{gg} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \text{ㅗ} & \\ \boxed{v_1} & \boxed{v_2} \\ \text{o} & \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \text{ㅊ} \\ \boxed{c_2} & \boxed{c_3} \\ \text{d} & \end{pmatrix}}_{c_o} \tag{31}$$

$$= \begin{pmatrix} \text{ㄲ} & \text{ㅗ} & \text{ㅊ} \\ \boxed{c_i} & \boxed{v_t} & \boxed{c_o} \\ \text{gg} & \text{o} & \text{d} \end{pmatrix} = \begin{pmatrix} \text{꽃} \\ \boxed{\text{syllable}} \\ \text{ggod} \end{pmatrix} \tag{32}$$

5.4. Initial consonant + compound vowel + optional consonant

$$\text{왔} = \underbrace{\begin{pmatrix} \text{ㅇ} \\ \boxed{c_1} \\ \varnothing \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \text{ㅗ} & \text{ㅏ} \\ \boxed{v_1} & \boxed{v_2} \\ \text{o} & \text{a} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \text{ㅅ} & \text{ㅅ} \\ \boxed{c_2} & \boxed{c_3} \\ \text{d} & \text{d} \end{pmatrix}}_{c_o} \tag{33}$$

$$= \begin{pmatrix} \text{ㅇ} & \text{ㅘ} & \text{ㅆ} \\ \boxed{c_i} & \boxed{v_t} & \underbrace{\boxed{c_o}}_{\text{compressed}} \\ \varnothing & \text{wa} & \text{dd} \end{pmatrix} = \begin{pmatrix} \text{왔} \\ \boxed{\text{syllable}} \\ \text{wadd} \end{pmatrix} \tag{34}$$

$$\text{닭} = \underbrace{\begin{pmatrix} \text{ㄷ} \\ \boxed{c_1} \\ \text{d} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \text{ㅏ} \\ \boxed{v_1} & \boxed{v_2} \\ & \text{a} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \text{ㄹ} & \text{ㄱ} \\ \boxed{c_2} & \boxed{c_3} \\ \text{l} & \text{g} \end{pmatrix}}_{c_o} \tag{35}$$

$$= \begin{pmatrix} \text{ㄷ} & \text{ㅏ} & \text{ㄺ} \\ \boxed{c_i} & \boxed{v_t} & \underbrace{\boxed{c_o}}_{\text{compressed}} \\ \text{d} & \text{a} & \text{lg} \end{pmatrix} = \begin{pmatrix} \text{닭} \\ \boxed{\text{syllable}} \\ \text{dalg} \end{pmatrix} \tag{36}$$

5.5. With special cases

$$\text{시} \overset{\text{special case}}{=} \underbrace{\begin{pmatrix} \text{ㅅ} \\ \boxed{c_1} \\ \text{s} \\ \text{sh} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \text{ㅣ} \\ \boxed{v_1} & \boxed{v_2} \\ & \text{i} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \boxed{c_2} & \boxed{c_3} \end{pmatrix}}_{c_o} \tag{37}$$

$$= \begin{pmatrix} \text{ㅅ} & \text{ㅣ} & \boxed{c_o} \\ \boxed{c_i} & \boxed{v_t} & \\ \text{s} & \text{i} & \\ \text{sh} & & \end{pmatrix} = \begin{pmatrix} \text{시} \\ \boxed{\text{syllable}} \\ \text{si} \times \\ \text{shi} \checkmark \end{pmatrix} \tag{38}$$

$$\underbrace{\text{빨}}_{\text{bbal}}\underbrace{\text{리}}_{\text{li}} \overset{\text{special case}}{=} \underbrace{\begin{pmatrix} \text{ㄹ} \\ \boxed{c_1} \\ \text{r} \\ \text{l} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \text{ㅣ} \\ \boxed{v_1} & \boxed{v_2} \\ & \text{i} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \boxed{c_2} & \boxed{c_3} \end{pmatrix}}_{c_o} \tag{39}$$

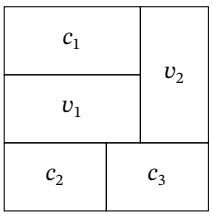
$$= \begin{pmatrix} \text{ㄹ} & \text{ㅣ} & \boxed{c_o} \\ \boxed{c_i} & \boxed{v_t} & \\ \text{r} & \text{i} & \\ \text{l} & & \end{pmatrix} = \begin{pmatrix} \text{리} \\ \boxed{\text{syllable}} \\ \text{ri} \times \\ \text{li} \checkmark \end{pmatrix} \tag{40}$$

$$\underbrace{\text{있}}_{\text{iss}}\underbrace{\text{어}}_{\text{uh}} \overset{\text{special case}}{\underset{\text{eq. (7)}}{=}} \underbrace{\begin{pmatrix} \text{ㅇ} \\ \boxed{c_1} \\ \varnothing \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \text{ㅣ} \\ \boxed{v_1}\ \boxed{v_2} \\ \text{i} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \text{ㅅ}\ \text{ㅅ} \\ \boxed{c_2}\ \boxed{c_3} \\ \text{d}\ \text{d} \\ \text{s}\ \text{s} \end{pmatrix}}_{c_o} \tag{41}$$

$$= \begin{pmatrix} \text{ㅇ} & \text{ㅣ} & \underbrace{\text{ㅆ}}_{\text{compressed}} \\ \boxed{c_i} & \boxed{v_t} & \boxed{c_o} \\ \varnothing & \text{i} & \underbrace{\text{dd}}_{\text{compressed}}\ \underbrace{\text{ss}}_{\text{compressed}} \end{pmatrix} = \begin{pmatrix} \text{있} \\ \boxed{\text{syllable}} \\ \text{idd} \times \\ \text{iss} \checkmark \end{pmatrix} \tag{42}$$

6. EXERCISES

Reminder.



c…consonant
v…vowel

Reminder: Syllable Block

$$\begin{pmatrix} \textbf{hangul letter} \\ \boxed{\text{position in Block}} \\ \text{default sound} \\ \text{special case} \end{pmatrix} \tag{43}$$

$$= \underbrace{\begin{pmatrix} \boxed{c_1} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \boxed{v_1}\ \boxed{v_2} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \boxed{c_2}\ \boxed{c_3} \end{pmatrix}}_{c_o} \tag{44}$$

$$= \begin{pmatrix} \boxed{c_i}\ \boxed{v_t}\ \boxed{c_o} \end{pmatrix} = \begin{pmatrix} \boxed{\text{syllable}} \end{pmatrix}_{\text{Answer:}} \tag{45}$$

6.1. Initial consonant + single vowel

$$\text{나} = \underbrace{\begin{pmatrix} \boxed{c_1} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \boxed{v_1}\ \boxed{v_2} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \boxed{c_2}\ \boxed{c_3} \end{pmatrix}}_{c_o} \tag{46}$$

$$= \begin{pmatrix} \boxed{c_i}\ \boxed{v_t}\ \boxed{c_o} \end{pmatrix} = \begin{pmatrix} \boxed{\text{syllable}} \end{pmatrix}_{\text{Answer: na}} \tag{47}$$

$$\text{미} = \underbrace{\begin{pmatrix} \boxed{c_1} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \boxed{v_1}\ \boxed{v_2} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \boxed{c_2}\ \boxed{c_3} \end{pmatrix}}_{c_o} \tag{48}$$

$$= \begin{pmatrix} \boxed{c_i}\ \boxed{v_t}\ \boxed{c_o} \end{pmatrix} = \begin{pmatrix} \boxed{\text{syllable}} \end{pmatrix}_{\text{Answer: mi}} \tag{49}$$

$$\text{개} = \underbrace{\begin{pmatrix} \boxed{c_1} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \boxed{v_1}\ \boxed{v_2} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \boxed{c_2}\ \boxed{c_3} \end{pmatrix}}_{c_o} \tag{50}$$

$$= \begin{pmatrix} \boxed{c_i}\ \boxed{v_t}\ \boxed{c_o} \end{pmatrix} = \begin{pmatrix} \boxed{\text{syllable}} \end{pmatrix}_{\text{Answer: gae}} \tag{51}$$

$$\text{너} = \underbrace{\begin{pmatrix} \boxed{c_1} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \boxed{v_1}\ \boxed{v_2} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \boxed{c_2}\ \boxed{c_3} \end{pmatrix}}_{c_o} \tag{52}$$

$$= \begin{pmatrix} \boxed{c_i}\ \boxed{v_t}\ \boxed{c_o} \end{pmatrix} = \begin{pmatrix} \boxed{\text{syllable}} \end{pmatrix}_{\text{Answer: nuh}} \tag{53}$$

$$\text{도} = \underbrace{\begin{pmatrix} \boxed{c_1} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \boxed{v_1}\ \boxed{v_2} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \boxed{c_2}\ \boxed{c_3} \end{pmatrix}}_{c_o} \tag{54}$$

$$= \begin{pmatrix} \boxed{c_i}\ \boxed{v_t}\ \boxed{c_o} \end{pmatrix} = \begin{pmatrix} \boxed{\text{syllable}} \end{pmatrix}_{\text{Answer: do}} \tag{55}$$

$$\text{그} = \underbrace{\begin{pmatrix} \boxed{c_1} \end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix} \boxed{v_1}\ \boxed{v_2} \end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix} \boxed{c_2}\ \boxed{c_3} \end{pmatrix}}_{c_o} \tag{56}$$

$$= \begin{pmatrix} \boxed{c_i}\ \boxed{v_t}\ \boxed{c_o} \end{pmatrix} = \begin{pmatrix} \boxed{\text{syllable}} \end{pmatrix}_{\text{Answer: geww}} \tag{57}$$

$$\text{효} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{58}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: hyo}} \tag{59}$$

6.2. Initial consonant + compound vowel

$$\text{왜} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{60}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: wae}} \tag{61}$$

$$\text{위} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{62}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: wi}} \tag{63}$$

$$\text{뭐} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{64}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: mwo}} \tag{65}$$

$$\text{의} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{66}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: ui}} \tag{67}$$

6.3. Initial consonant + single vowel + optional consonant

$$\text{산} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{68}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: san}} \tag{69}$$

$$\text{물} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{70}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: mool}} \tag{71}$$

$$\text{강} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{72}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: gang}} \tag{73}$$

$$\text{옷} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{74}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: od}} \tag{75}$$

$$\text{낮} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{76}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: nad}} \tag{77}$$

6.4. Initial consonant + compound vowel + optional consonant

$$\text{원} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{78}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: won}} \tag{79}$$

$$\text{괜} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{80}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: gwaen}} \tag{81}$$

$$\text{봤} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{82}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: bwadd}} \tag{83}$$

$$\text{읽} = \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{84}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: ilg}} \tag{85}$$

6.5. With special cases

$$\underbrace{\text{몰}}_{\text{mol}}\underbrace{\text{라}}_{\text{??}} \overset{\text{special case}}{=}_{\text{eq. (5)}} \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{86}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: la}} \tag{87}$$

$$\underbrace{\text{맞}}_{\text{??}}\underbrace{\text{아}}_{\text{a}} \overset{\text{special case}}{=}_{\text{eq. (8)}} \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{88}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: maj}} \tag{89}$$

$$\underbrace{\text{옷}}_{\text{??}}\underbrace{\text{이}}_{\text{i}} \overset{\text{special case}}{=}_{\text{eq. (7)}} \underbrace{\boxed{c_1}}_{c_i} + \underbrace{\boxed{v_1}\ \boxed{v_2}}_{v_t} + \underbrace{\boxed{c_2}\ \boxed{c_3}}_{c_o} \tag{90}$$

$$= \left(\boxed{c_i}\ \boxed{v_t}\ \boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right)_{\text{Answer: os}} \tag{91}$$

6.6. Whole words

$$\underbrace{\text{우유}}_{\text{milk}} = \left(\boxed{\text{syllable}_1}\ \boxed{\text{syllable}_2}\right)_{\text{Answer: oo-you}} \tag{92}$$

$$\underbrace{\text{사랑}}_{\text{love}} = \left(\boxed{\text{syllable}_1}\ \boxed{\text{syllable}_2}\right)_{\text{Answer: sa-rang}} \tag{93}$$

$$\underbrace{\text{친구}}_{\text{friend}} = \left(\boxed{\text{syllable}_1}\ \boxed{\text{syllable}_2}\right)_{\text{Answer: chin-goo}} \tag{94}$$

$$\underbrace{\text{김치}}_{\text{fermented cabbage}} = \left(\boxed{\text{syllable}_1}\ \boxed{\text{syllable}_2}\right)_{\text{Answer: gim-chi}} \tag{95}$$

$$\underbrace{\text{한국}}_{\text{Korea}} = \left(\boxed{\text{syllable}_1}\ \boxed{\text{syllable}_2}\right)_{\text{Answer: han-goog}} \tag{96}$$

$$\underbrace{\text{신발}}_{\text{shoe}} = \Big(\boxed{\text{syllable}_1} \quad \boxed{\text{syllable}_2} \Big) \qquad (97)$$

Answer: shin-bal

7. LETTER INDEX

Each letter is followed by the number of the equation where it is described. Click a number to jump there.

Consonants

ㄱ (3)	ㄲ (3)	ㄴ (4)	ㄷ (4)	ㄸ (4)	ㄹ (5)	ㅁ (6)
ㅂ (6)	ㅃ (6)	ㅅ (7)	ㅆ (7)	ㅇ (9)	ㅈ (8)	ㅉ (8)
ㅊ (8)	ㅋ (3)	ㅌ (4)	ㅍ (6)	ㅎ (9)		

Vowels

ㅏ (15)	ㅐ (17)	ㅑ (15)	ㅒ (17)	ㅓ (16)	ㅔ (18)	ㅕ (16)
ㅖ (18)	ㅗ (12)	ㅘ (20)	ㅙ (20)	ㅚ (20)	ㅛ (12)	ㅜ (13)
ㅝ (21)	ㅞ (21)	ㅟ (21)	ㅠ (13)	ㅡ (11)	ㅢ (19)	ㅣ (14)

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